DE–4095

Sub. Code
101

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Convert the following :

   (a) $(1234.56)_{10} = (?)_2$

   (b) $(1234.56)_8 = (?)_{10}$

   (c) $(1234.56)_{16} = (?)_{10}$

   (d) $(110101.1101)_2 = (?)_{10}$

2. Describe the different types of complements.

3. State various Boolean laws.

4. Describe Decoder with neat diagram.

5. Write short notes on ROM.

6. Explain the organization of magnetic disk memory.

7. Draw the circuit of a shift register and explain the operations in detail.

8. Explain any two peripheral devices.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about the Karnaugh's map simplification with neat diagram.

10. Convert the following expressions to sum-of-products form:

(a) (A+B) (B+C) (A+C)

(b) (A+C) (A+B+C) (A+B)

(c) (A+C) (AB+AC) (AC+B)

11. Explain Multiplexer and Demultiplexer with neat diagram.

12. Explain the functionality of Half adder and full adder.

13. What is a counter? Design a mod-5 counter with J-K flip flop.

14. What is interrupt? Explain in detail about interrupt Priorities.

15. Explain the block diagram of an I/O interface unit.

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DE–4096

Sub. Code
102

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

C AND DATA STRUCTURES

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is type? Explain the difference type of data types with range.

2. What are various conditional statements? Explain the difference between nested if and switch case statement by taking suitable examples.

3. What is multidimensional array? Write a 'C' language program to find transpose of matrix by using the concept of functions.

4. What are header files? What is need of header files in 'C' programming?

5. Explain how 'While' statement is different from 'do' while 'statement' with suitable example program.

6. What is structure? How it is different from union?

7. Define Stack and Queue operation? Explain functions of stack and Queue.

8. What is binary tree? Solve the mathematical expressions (A+B)*C using binary tree structure.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss in detail about call by value and call by reference with suitable example program.

10. What is file? Explain various function used for processing a file in 'C' language.

11. What is storage classes? Explain different types of storage classes.

12. Explain the role of preprocess directives in 'C' programming with suitable example.

13. Explain various string handling functions in C language with example program.

14. What is mean by Recursion? Explain with suitable example program.

15. What is linked list? Explain different types of linked list.

DE–4097

Sub. Code
103

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is a Data Model? Explain.

2. Give the usage of the Rename operation with an example.

3. Consider the following relation:

   EMP (ENO, NAME, DATE_OF_BIRTH, SEX, DATE_OF_JOINING, BASIC_PAY, DEPT) Develop an SQL query that will find and display the average BASIC_PAY in each DEPT.

4. Define Primary Key and Foreign Key with table explanations.

5. Give a brief description on DML commands.

6. Explain the overview of the Oracle Architecture.

7. Discuss in details the operators, SELECT, PROJECT and UNION with suitable examples.

8. Explain Joining Function with an examples.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss in detail about Data Base System Architecture with neat diagram.

10. Consider the following six relations for an Order-processing

Database Application in a Company:

CUSTOMER (CUSTNO, CNAME, CITY)

ORDER (ORDERNO, ODATE, CUSTNO, ORD_AMT)

ORDER_ITEM (ORDERNO, ITEMNO, QTY)

ITEM (ITEMNO, ITEM_NAME, UNIT_PRICE)

SHIPMENT (ORDERNO, ITEMNO, WAREHOUSENO,

SHIP_DATE)

WAREHOUSE (WAREHOUSENO, CITY)

Here, ORD_AMT refers to total amount of an order; ODATE is the date the order was placed; SHIP_DATE is the date an order is shipped from the warehouse. Assume that an order can be shipped from several warehouses. Specify the foreign keys for this schema, starting any assumptions you make.

11. Discuss the terms Distributed Database and Client/ Server Database.

12. Explain Data Independence and its types with examples.

13. Create two table, 'University' and 'Student' using Join function to join both the tables entity. And also use the COPY and DROP commands.

14. Explain about Discretionary access control based on GRANT and REVOKE privileges.

15. Explain the following with examples

    (a) DML

    (b) DCL

    (c) Embedded SQL.

DE–4098

Sub. Code
104

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

OPERATING SYSTEMS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Mention the Objectives and Functions of an operating system.

2. Define Mutual exclusion and explain.

3. What is Deadlock? What are the schemes used in operating system to handle deadlock?

4. Differentiate a Page from a Segment.

5. Explain the Two level and Three level directory.

6. Define Disk Space Management. Explain with example.

7. Explain the various File access method.

8. What is the need for Disk Scheduling? Explain the Swap-Space management.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the purpose and importance of System calls in details with example.

10. What is meant by Critical Section problem? Give a solution based on Bakery algorithm.

11. Explain Banker's deadlock – avoidance algorithm with illustration.

12. Explain Memory Management.

13. Describe the Hierarchical paging techniques for structural page table.

14. Explain the FIFO, optimal and LRU page replacement algorithm?

15. Describe the IPC in Client-Server system.

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DE–7045

Sub. Code
105

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

DATA STRUCTURE AND USING C – LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by the lot system.

Each question carries 50 marks.

1. (a) Write a C program to find the roots of a quadratic equation.

$$ax^2 + bx + c = 0$$

(b) Write a C program to implement push() and pop() operations on stack.

---------- Cut here ----------

2. (a) Write a C program to find a given word is palindrome or not.

(b) Write a C program to implement insert( ) and delete( ) operations on queue using array.

3. (a) Write a C program to implement read write and append mode using file.

   (b) Write a C program to evaluate to your own mathematical expression using stack operation.

---------- Cut here ----------

4. (a) Write a C program to read 20 values to an array variable and to perform ascending order, descending order, average, maximum and minimum value of given array.

   (b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

5. (a) Write a program in C to search an element using linear and binary search.

   (b) Write a C program to sort a set of elements using insertion sort.

DE–7046

Sub. Code
106

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system

Break-up of Marks

Record note book	:	5
Algorithm/Flow chart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

---------- Cut here ----------

1. A Salary statement contains Name, Basic–pay, Allowances, Total Deduction (includes IT), Gross pay, Net-.pay.

Dearness Allowance	=	58% of basic pay
Gross-pay	=	Basic–pay + D.A
Deduction	=	P.F + LOAN AMT

Income Tax is calculated on the basis of Annual income with the following conditions.

Annual Salary	Income Tax
<= 1 Lakh	Nil
>1 Lakh but <= 1.5 Lakh	10%
>1.5 Lakh but <= 3 Lakh	20%
>3 Lakh and above	30%

Total deduction = Deduction + Income Tax

Write a program in SQL to prepare Salary report for 25 employees.

---------- Cut here ----------

2. Write a program in SQL for Hospital Billing System with the following fields:

Patient no., Patient name, Age, Attending Doctor, Patient- type(IN/OUT), Consulting charge Blood test charge, X-Ray charges, other test charges and total fee.

Prepare reports with the following conditions:

(a) The patients, undergone blood test

(b) The patients, who taken X-rays

(c) The patients, who belong to In-patient category

The common fields to be included in the above mentioned reports: Patient No, Patient name, Age, Charges and Total charges.

3. A hospital maintains blood donors' record information in a file. The file contains the following fields:

   Donor number, Donor name, Donor age, Donor Address, Pin code, Place of birth, Blood group (A,B,AB &O).

   Write a program using Oracle to print out the number, name and address of the donors for the following categories :

   (a) Blood donors having AB blood group.

   (b) Blood donors in the Age group between 16 to 25.

   (c) Female donors with 'A' blood group in the age between 18 and 25.

---------- Cut here ----------

4. Write a SQL program for the Canteen Maintenance with two tables namely MENU and BILL.

   The MENU table contains the following fields :

   ITEM , UNIT-RATE.

   The following items are available in the canteen :

   Tea, Coffee and Cool-drinks

   The BILL table contains the following fields :

   EMPNO, EMPNAME, DATE OF ISSUE, ITEM1, NO.OF TOKENS ISSUED FOR ITEM1, RATE1, ITEM2, NO.OF TOKENS ISSUED FOR ITEM2, RATE2, ITEM3, NO.OF TOKENS ISSUED FOR ITEM3, RATE3, TOTAL AMOUNT.

   Note : An employee is allowed to get the token only once in a month.
   RATE = UNIT-RATE * NO. OF TOKENS FOR PARTICULAR ITEM TOTAL AMOUNT =RATE-1 + RATE-2 + RATE-3

Prepare a report in the following format :

S.no.	Empno	Empname	Purchased	Rate
1	–	–		
2	–	–		
3	–	–		
			Total amount	

---------- Cut here ----------

5. An airline reservation database contains the reservation table and personal table.

Reservation table contains the following fields:

Flight number, Class, Passenger Name, Seat Number

Personal table contains the following fields:

Passenger name, Sex, Age, Martial status, Nationality

Write a program in ORACLE to prepare the following reports :

(a) List the passenger names with seat number according to class and flight numbers

(b) Total number of married female passengers in a particular flight

(c) List out male passengers in the age between 18 and 25 for all flights.

(d) List passengers travelling in First class.

6. A News paper vendor buys newspapers on wholesale from a Distributor for Rs. 4 and sells them in retail for Rs. 4.50. At the end of the day the unsold papers are returned to the distributor for a Re. 1 rebate per paper. Write a program using Oracle to prepare a port for the newspaper vendor in the following format with 9 weeks data :

Week	Bought	Sold	Return	Profit	Loss
1	100	50	XX	XXX.XX	XXX.XX
2	75	60	XX	XXX.XX	XXX.XX
–	–	–	–	———	———
Total					

---------- Cut here ----------

7. An examination has been conducted to a class of 55 students and the following are given as inputs :

Three test marks, Register No and Candidate Name.

Write a SQL program to do the following:

Assign a letter grade to each student based on the average score and list Register Number, Average score and grade. The minimum pass for each subject is 50.

The grading system:

Average Score	Grade
90–100	A
75–89	B
60–74	C
50–59	D
0–49	F-fail

8. Write a PLSQL program to prepare a mark statement of a student using Student Table and Mark table. Assume your own data.

---------- Cut here ----------

9. Write a PL/ SQL program to prepare an electricity bill. Assume your own data.

---------- Cut here ----------

10. Create a table which contains phone number, name and address. Write a PL/SQL program to search for address using phone number.

DE–4099

Sub. Code
201

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is POP? Discuss its features. Describe how data are shared by functions in procedure-oriented programs?

2. Write a program in C++ to find the prime factors of a given number.

3. What is inline function? When will you make a function inline and why?

4. Describe the concept of operator overloading using member function.

5. Give and explain the two ways for opening a file in C++.

6. What do you mean by dynamic binding? How is it useful in OOP?

7. Illustrate the working of destructor function with any three special characteristics of destructors.

8. Develop an application for pay slip preparation in C++ with multiple inheritance.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Define Manipulators. Explain the functions of any three standard library manipulators with suitable examples. Mention any three applications of OOP.

10. Implement mark statement with grades by creating list of students in a class using structures. (Assume grades for range of marks on your own).

11. Define virtual function. Explain the usage and importance of virtual function with an example. How does it differ from other methods?

12. Define a class "string" with members to initialize and determine the length of the string. Overload the operators '+' and '+=' for the class "string".

13. Give the functions for manipulations of file pointer. Describes the various approaches by which we can detect the end of file condition.

14. Explain and compare various types of inheritance available in C++.

15. Implement Fibonacci series in C++ with recursion. Also explain this recursion's internal operation with the desired data structure.

DE–4100

Sub. Code
202

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

UNIX AND SHELL PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the following commands :
   (a) who
   (b) echo
   (c) cd
   (d) pwd.

2. Discuss about Vi editor and its basic operations.

3. Explain about switch-case statement in shell programming with an example.

4. Explain Unix system calls.

5. Explain the declaration of structure and accessing structure with suitable example.

6. Discuss about shell and execution of shell script with an example.

7. How to access the environment variables in Unix?

8. Write short notes on standard input and output in Unix.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Describe the architecture of UNIX operating system, with all its modules.

   (b) Discuss briefly UNIX file system structure.

10. What are the basic functions of shell? Explain the different types of shells used in UNIX OS.

11. (a) Explain about the useful permission setting for a file and a directory.

    (b) Write the syntax for chmod and unmask.

12. (a) What are filters? Give examples for filters.

    (b) Write a shell program to print the given number is prime or not.

13. (a) What is the difference between built-in and utility commands?

    (b) Explain parent, child and sub shells.

14. Explain about IF, WHILE and FOR loop structure with suitable examples.

15. Write short notes on :

    (a) Pattern matching

    (b) Computation and string handling.

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DE–4101

Sub. Code
203

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

COMMUNICATION SKILLS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define Communication. Also write its importance.

2. What are the barriers to communication?

3. Write a note on the process of presenting reports.

4. What are the different types of body language?

5. Explain the negotiation techniques.

6. How are communication skills important for seminars?

7. Give an account on designing.

8. Write a note on postscript.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the objectives of minutes and communication aids.

10. Trace the techniques of reading communication skills.

11. Describe the importance of mock group discussion.

12. What are the tools used for personality identification in body language? Explain.

13. Explain the managing process of negotiation.

14. Examine the Union's role in communication.

15. Distinguish upward and downward communication.

DE-4102

Sub. Code
204

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

COMPUTER GRAPHICS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various Graphical Output devices.

2. Write Bresenham's algorithm to draw a line.

3. Discuss the two-dimensional transformations.

4. What is meant by viewport? Explain with a suitable example.

5. Write short notes on Mirror reflection.

6. Discuss the Parallel projection.

7. Discuss the Polygon clipping in three-dimensional view.

8. Write a note on Command languages.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the Spline curves with examples.

10. Describe the region filling technique.

11. Illustrate the two-dimensional transformations with matrix notation.

12. Discuss the Sutherland Hodgman algorithm for polygon clipping.

13. Explain the composite 3-D transformations.

14. Explain the Hidden surface algorithm.

15. Describe the depth buffer algorithm.

DE–7047

Sub. Code
205

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

C++ AND GRAPHICS LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Record : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

Examiner has to choose ONE question for each candidate by lot system.

Each sub-division carries 50 marks.

1. (a) Create a base class shape, which contains the common element of circle from which you derive a class called ball. Include two member functions set() to set coordinates and draw() which draws the circle. Write a main() program to exercise this class using graphic-mode graphic functions.

   (b) Write a C++ program to draw a straight line using simple DDA technique.

---------- Cut here ----------

2. (a) Write a C++ program to draw a straight line using Bresenham's line drawing algorithm.

   (b) Write a C++ program to sketch a triangle.

3. (a) Develop a program using functions to perform the following :
   (i) Draw a triangle at 90°.
   (ii) Rotate it through 90 degree.
   (iii) Translate to another position.
   (b) Fill the triangle drawn in (a) with read Green.

---------- Cut here ----------

4. (a) Create a base class shape, which contains the common element of circle from which you derive a class called bail. Include two member functions set() to set coordinates and draw() which draws the circle. Write a main() program to exercise this class using graphic-mode graphic functions.
   (b) Write a program in C++ to draw a polygon.

---------- Cut here ----------

5. (a) Write a C++ program to draw a Circle, and fill with magenta colour.
   (b) Write a C++ program to sketch a Hexagon.

---------- Cut here ----------

6. Develop a program to perform the following :
   (a) Draw a triangle ABC, translate it to a distance and rotate it through an angle clock-wise.
   (b) Write a program in C++ to draw a polygon.

---------- Cut here ----------

7. (a) Write a program in C++ to draw Concentric circle with four objects.
   (b) Draw a ball, and make animation effect on it.

8. (a) Write a C++ program to perform graphical transformation (Scaling, Rotation, Translation) of a two dimensional object.

(b) Draw a ball, and make animation effect on it.

---------- Cut here ----------

9. (a) Write a C++ program to draw triangle using SHADING technique.

(b) Write a program in C++ to clip a part of a line.

---------- Cut here ----------

10. (a) Draw a Circle using Bresenham's algorithm, and fill with red colour.

(b) Make animation effect on the Circle drawn in (a).

---------- Cut here ----------

11. (a) Develop a program using functions to perform the following :

(i) Draw a rectangle

(ii) Rotate it through 90 degree

(iii) Translate to another position.

(b) Write a program in C++ to clip a part of a Polygon.

---------- Cut here ----------

12. (a) Draw a ball, and make animation effect on it.

(b) Draw a triangle ABC, translate it to a distance and then rotate it through an angle Clock-wise.

13. (a) Draw a Circle, and make animation effect on it.

(b) Develop a program using functions to perform the following :

(i) Draw a Square.

(ii) Rotate it through 45 degree.

(iii) Translate to another position.

---------- Cut here ----------

14. (a) Write a C++ program to draw triangle using SHADING technique.

(b) Write a program in C++ to clip a part of a Polygon.

---------- Cut here ----------

15. (a) Create a base class shape, which contains the common element of circle from which you derive a class called ball. Include two member functions set() to set coordinates and draw() which draws the circle. Write a main() program to exercise this class using graphic-mode graphic functions.

(b) Draw a triangle ABC, translate it to a distance and then rotate it through an angle Clock-wise.

DE–7048

Sub. Code
206

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

UNIX AND SHELL PROGRAMMING LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

One question should be selected by the examiner and given to the candidates by lot system.

1. (a) Write a shell script that takes a login name as command – line argument and reports when that person logs in.

   (b) Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.

---------- Cut here ----------

2. (a) Develop an interactive script that ask for a word and a file name and then tells how many times that word occurred in the file.

   (b) Write a shell script to perform the following string operations :

      (i) To extract a sub-string from a given string.

      (ii) To find the length of a given string.

3. (a) Write a shell program which periodically monitors the disk for the existence of a tile and executes the program once the file has been located.(using UNTIL statement)

   (b) Write a shell script that displays a list of all the files in the current directory to which the user has r3ad, write and execute permissions.

---------- Cut here ----------

4. (a) Write a shell script that accepts one or more file name as arguments and converts all of them to uppercase, provided they exist in the current directory.

   (b) Write a shell script that takes a command – line argument and reports on whether it is a directory, file, or something else.

---------- Cut here ----------

5. (a) A hospital maintains patient details. Write a shell program to list the following :

      (i) Patients in a blood group wise.

      (ii) Patients in age between 20 to 30.

      (iii) Exit.

   (b) Write a program for four-function calculator by using yace command.

---------- Cut here ----------

6. (a) Write a shell program for the following :

      (i) List the files in date order

      (ii) Sort the files in descending order.

   (b) Write a shell program to check whether given number is even or odd using positional parameters.

7. (a) Write a shell script that computes the gross salary of a employee according to the following rules :

   (i) If basic salary is < 1500 then HRA = 10% of the basic and DA 90% of the basic.

   (ii) If basic salary is >=1500 then HRA = Rs. 500 and DA = 98% of the basic.

   The basic salary is entered interactively through the key board.

   (b) Write a shell program to reverse the given string.

---------- Cut here ----------

8. (a) Write and execute the UNIX Commands for performing the following :

   (i) To create a directory

   (ii) To compare two files and show the differences.

   (iii) To pipe the output of "who" command to the "sort" command.

   (iv) To change the ownership of a file

   (v) To save the output in a file and display the same on the VDU.

   (b) Write a shell script to display user name, current directory, current date and time.

---------- Cut here ----------

9. (a) Write a shell program to perform the following :

   (i) Read any text file

   (ii) Print its contents without using 'cat' / 'pr commands.

   (b) Write a shell program to concatenate and copy two strings without using built-in functions.

10. (a) Write a shell program to check if a number is positive, negative or zero.

(b) Write a shell program to find the following objects :

(i) Null string

(ii) Not a null string.

DE–4103

Sub. Code
301

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Third Semester

DISCRETE MATHEMATICS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Prove $(P \to Q) \Leftrightarrow (\neg P \vee Q)$.

2. Show that $R \to S$ can be derived from the premises $P \to (Q \to S)$, $\neg R \vee P$ and Q.

3. Define power set. Find the power set of $S = \{X, Y, Z\}$.

4. If $X = \{1, 2, 3, 4\}$ and $R = \{\langle X, Y \rangle \mid X > Y\}$. Draw the graph of R.

5. Show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

6. Define the following with an example :

   (a) Sub semigroup

   (b) Submonoid.

7. Prove that the ring of even integers is a subring of the ring of integers.

8. Define the following with an example :

   (a) Weighted graph

   (b) Path of a graph

   (c) Connected graph

   (d) Adjacency matrix of a graph.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Show that the following are equivalent formulas

   (i) $P \vee (P \wedge Q) \Leftrightarrow P$

   (ii) $P \vee (\neg P \wedge Q) \Leftrightarrow P \vee Q$

   (b) Obtain the principal conjuctive normal form of the following formula $P \vee (\neg P \rightarrow (Q \vee (\neg Q \rightarrow R)))$.

(8 + 7)

10. (a) Prove that $(\exists x)(P(x) \wedge Q(x)) \Rightarrow (\exists x) P(x) \wedge (\exists x) Q(x)$.

   (b) Show the following by constructing derivations :

   $(x)P(x) \rightarrow Q(y) \wedge R(x)$. (8 + 7)

11. (a) If relations R and S are reflexive, symmetric and transitive show that $R \cap S$ is also reflexive, symmetric and transitive.

   (b) If A be the set of factors of a particular positive integer m and let $\leq$ be the relation divides i.e., $\leq = \{\langle x, y\rangle \mid x \in A \wedge y \in A \wedge (x \text{ divides } y)\}$. (8 + 7)

12. (a) If $f(x) = x + 2, g(x) = x - 2$, and $h(x) = 3x$ for $x \in R$ where R is the set of real numbers. Find $g \circ f; f \circ g; f \circ f$.

(b) Show that $f : X \to Y$ is one-to-one iff any proper subsets of X are mapped into proper subsets of Y. (8 + 7)

13. (a) Prove that the order of a subgroup of a finite group divides the order of the group.

(b) If $\langle G, * \rangle$ is an abelian group, then for all $a, b \in G$ show that $(a * b)^n = a^n * b^n$. (8 + 7)

14. (a) If $\langle G, * \rangle$ and $\{H, \Delta\}$ be groups and $g . G \to H$ be a homomorphism then prove that the Kernal of g is a normal subgroup.

(b) Show that every subgroup of a cyclic group is normal. (8 + 7)

15. In a simple digraph $G = \langle V, E \rangle$, prove that every nodes of the digraph lies in exactly one strong component.

DE-4104

Sub. Code
302

DISTANCE EDUCATION

M.C.A/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Third Semester

COMPUTER NETWORKS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the communication model with suitable diagram.

2. Discuss the various techniques in error detection.

3. Explain in detail packet switching techniques.

4. How the connections are established in the transport layer? Discuss.

5. Explain the digital signal encoding techniques.

6. Write a short note on Internet Multiplexing.

7. Explain the various layers in ATM.

8. How do the Domain Name Service provided? Explain

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss about the network standardization.

10. Explain the HDLC protocol with help of a frame format.

11. Explain about Internet Control Message Protocol.

12. Illustrate the TCP/IP model and the functions of various layers, in detail.

13. Explain the components of the SNMP management model. What are the basic functions supported by Email systems?

14. Explain in detail the DES algorithm with its merits and demerits.

15. Explain congestion avoidance techniques in detail.

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DE–4105

Sub. Code
303

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Third Semester

SOFTWARE ENGINEERING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define software metrics. Mention and narrate the attributes of effective software metrics.

2. Explain chief programmer and hierarchical programming team structure with their advantages.

3. Illustrates the format of a time-line chart. Give the different ways to track the schedule.

4. What are the good components of software requirement specification? How to write a software requirements specification (SRS) document?

5. Describe the concepts to be followed in module levels. Compare architectural design with interface design.

6. What are the size factors in software engineering? Explain the features of any two of them.

7. Describe software risks. Give the procedure for identifying the software risks.

8. List the risk components and explain how to assess the overall project risk.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Mention the software development process models and explain with an example of software projects that would be amenable to the spiral model.

10. Explain any two software cost estimation techniques.

11. What is risk projection? How to develop a risk table and to assess risk impact?

12. Give the necessity of prototyping in software requirement specification. List and explain any one software requirement specification techniques.

13. Describe the various testing strategies to be followed on developing a software with example.

14. Explain the various formats in project structure.

15. Describe three major categories of software engineering resources.

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DE–4106

Sub. Code
304

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Third Semester

VISUAL PROGRAMMING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. (a) What is the function of a Timer control?

   (b) What are the various Control Structures supported by Visual Basic?

2. How a DLL can be created using MFC? Explain in detail.

3. (a) What is Status bar? Explain the properties and methods.

   (b) What are the differences between Linked objects and Embedded objects?

4. (a) Implement VB program to check whether an element is present in an array or not by using required control.

   (b) Implement VB program to generate the Fibonacci series.

5. Explain how exception handling is done using MFC.

6. What are Multiple views? How are Multiple views used? How a New view is created?

7. (a) Write down the difference between structured programming and object oriented programming.

   (b) Explain how a window is created using MFC.

8. What are the controls that provide choices to the users in Visual Basic? Explain in detail.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Develop a program using SDK to implement an user interactive application through Mouse device.

   (b) Narrate the types of function calls associated with MDI.

10. (a) List and explain the step by step procedure to develop and run a project in VC++.

    (b) How a message box can be added? Explain it with the help of MFC.

11. (a) With suitable example, explain all looping statements in Visual Basic.

    (b) Write short notes on Events and Modules.

12. List any five numeric functions and any five string functions with examples.

13. (a) Describe about the Database management with ODBC in Visual Basic.

    (b) What is meant by Debugging? Explain.

14. Discuss the uses of following with examples :

    (a) CArchive

    (b) CWnd

    (c) CEdit

    (d) CDialog

    (e) Object.

15. (a) Explain how to give connectivity to a data source using VC++ in detail with suitable examples.

    (b) Write short notes on App Wizard in VC++.

DE–7049

Sub. Code
305

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Third Semester

NETWORK LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by the lot system.

Each subdivision carries 50 marks.

1. (a) Write a program in java to establish a communication between client and server.

   (b) Write a java program to find factorial of a given number.

---------- Cut here ----------

2. (a) Write a java program to read a file from a remote system.

   (b) Write a java program, which compute cube of a given number.

3. (a) Write a java program to display date and time from local host to server using TCP.

   (b) Write a java program for UDP chat server between multiple clients and a single server.

---------- Cut here ----------

4. (a) Write a java program to add two numbers using RMI client, server technique.

   (b) Write a java program to create client server communication using UDP sockets.

---------- Cut here ----------

5. (a) Write a java program to find factorial of a given number using RMI client, server technique.

   (b) Write a java program to display the users and their IP address.

---------- Cut here ----------

6. (a) Write a java program to implement Fibonacci sequence for a given number on a remote host and obtain its value from a client.

   (b) Write a java program to manipulate the IP address.

7. (a) Write a java program for transferring a file using TCP.

   (b) Write a java program, which helps to set Date and Time in client-server.

---------- Cut here ----------

8. (a) Write a java program to ENCRYPT a given word from server and display the same word in the client.

   (b) Write a java program to subtract two numbers using RMI client, server technique.

DE–7050

Sub. Code
306

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2018.

Third Semester

VISUAL PROGRAMMING LAB

(2010 Academic Year onwards)

Time : Three Hours Maximum : 100 Marks

Examiner should select and give ONE question to each candidate by lot system.

Break-up of marks

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. (a) Develop a VB project using control array that creates a scientific Calculator with appropriate command buttons. Include the following capabilities for the calculator: +, –, *, /, %, power, square root, square and log (base 10).

   (b) Write a Visual C++ win32 application program using MFC that creates a new font.

2. (a) Develop a VB project using Ax DLL or EXE add a class model that would perform the following functions :

   (i) Test whether the given number is perfect or not

   (ii) Test whether the given number is Armstrong or not

   (iii) Find the factorial of a given number

   (iv) Find the sum of digits

   (b) Write a Visual C++ win32 application program using MFC that fills the background of the client area with a bitmap.

---------- Cut here ----------

3. (a) Develop a VB project using built in Ax control(Rich Text Box), develop the windows NOTEPAD like editor with File and Edit menus and also display the floating menu whenever necessary.

   (b) Write a Visual C++ win32 application program using MFC that displays the status of ALT, CTRL, SHIFT, NUM LOCK and SCROLL LOCK keys.

4. (a) Develop a VB project using Active X control to create a Textbox that accepts only a numeric value. Provide the following properties for the text box Back color, Fore color and Text.

   (b) Write a Visual C++ win32 application program using MFC that displays a message "Hello Good Morning!" wherever the user clicks the mouse button on the client area.

---------- Cut here ----------

5. (a) Develop a VB project that stores 10 employee records with fields EMPNO, NAME, AGE, SEX and SALARY in an array. Display data fields in text boxes and provide command buttons to move to desire record.

   (b) Write a Visual C++ win32 application program using MFC that creates a list box and displays name of the states in India.

---------- Cut here ----------

6. (a) Develop a VB project for commercial bank operations using SB account database with the following features :
   (i) ADD NEW ACCOUNT
   (ii) DEPOSIT AMOUNT
   (iii) WITHDRAW AMOUNT (with minimum balance condition)

   (b) Write a Visual C++ win32 application program using MFC that creates two push buttons OK and CANCEL on the client area. Buttons should respond to user click over them and display appropriate message.

7. (a) Develop a VB project that receives a foreign currency value selected from a list box and converts it into equivalent Indian rupees(e.g USD 42.45, Sterling 71.30, D.Mark 25.52, SW Franc 31.58, SaudiRiyal 11.40, French Franc 7.60, UAE Dhiram 11.55, Kuwait Dhinar 140.56).

   (b) Write a Visual C++ win32 application program using MFC that displays current mouse coordinates in status bar.

---------- Cut here ----------

8. (a) Develop a VB project that receives a year number from a text box and month name from list box and displays number of days in the given month. Take care of leap years. Use Lost Focus event for list box.

   (b) Write a Visual C++ win32 application program using MFC that creates a list box and displays name of the states in India.

DE–4107

Sub. Code
401

DISTANCE EDUCATION

M.C.A. /M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define accounting. Explain its scope.

2. Prepare a cost sheet form the following data :

	Rs.
Direct materials	20,000
Direct wages	10,000
Factory overheads	5,000
Selling overheads	3,000
Office expenses	2,000
Opening stock of raw materials	5,000
Closing stock of raw materials	7,000
Opening stock of work in progress	3,000
Closing stock of work in progress	2,000
Sales	1,20,000

3. From the following balances you are required to calculate cash from operations :

	31-12-2014 Rs.	31-12-2015 Rs.
Debtors	50,000	47,000
Bills Receivable	10,000	12,500
Creditors	20,000	25,000
Bills payable	8,000	6,000
Outstanding expenses	1,000	1,200
Prepaid expenses	800	700
Accrued income	600	750
Income received in advance	300	250
Profit made during the year	-	1,30,000

4. From the following data, compute current assets, Quick assets and Stock :

Current liabilities	Rs. 9,00,000
Current ratio	2.5
Acid test ratio (without prepaid expenses)	2

5. From the following data, determine the break-even point:

Fixed overhead : Rs. 25,000

Variable cost : Rs. 10 per unit

Selling price : Rs. 15 per unit

What should be the sales volume to make a profit of Rs. 25,000?

6. Briefly explain the various types of functional budgets.

7. Rajan supplies components to I.C.F on contract basis. For the year 2000, he agrees to supply 20,000 units at Rs. 80 per unit. The following were his costs during 1999 for supply of 15,000 units.

Material Rs. 8 per unit

Wages Rs. 22 per unit

Overheads Rs. 20 per units (60% fixed)

Other expenses Rs.5 (per unit) (100% fixed)

Prepare budget for the year 2000, showing clearly the budgeted profit.

8. State and explain the goals of financial management.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. The following trial balance was taken from the books of a sole trader for the year ended 31st 9 March 2013 :

Account Balances	Debit (Rs.)	Credit (Rs.)
Purchase and sales	92,300	1,90,300
Carriage inwards	5,200	
Drawings	5,000	
Rent, rates and insurance	4,500	
Postage	3,000	
Stationery	2,700	

Capital		59,400
Machinery	55,000	
Buildings	45,000	
Furniture and fittings	15,000	
Debtors and creditors	14,500	13,200
Commission received		7,000
Opening stock	1,000	
Cash at bank	2,500	
Cash in hand	1, 800	
Discounts	250	3,920
Bad debt	800	
Salaries and wages	22,120	
Advertising	1,850	
Carriage outwards	1,200	
Returns in and out	800	700
Total	2,74,520	2,74,520

Consider the following points at 31st March 2013:

(a) The closing stock was valued at Rs. 12,400.

(b) Rent, rates and insurance outstanding by Rs. 500.

(c) Commission received in advance is Rs. 1,000.

(d) Salaries and wages owing Rs. 880.

(e) Advertisement expense is prepaid by Rs.350.

(f) Carriage inwards payable Rs.800.

From the above information, you are required to prepare trading, profit and loss account, and balance sheet at 31st March 2013.

10. Balance sheets of M/s. Black and White as on 1.1.2012 and 31.12.2012 were as follows :

Liabilities	1.1.2012	31.12.2012	Assets	1.1.2012	31.12.2012
Creditors	40,000	44,000	Cash	10,000	7,000
Mr. Whites loan	25,000	–	Debtors	30,000	50,000
Loan from bank	40,000	55,000	Stock	35,000	25,000
Capital	1,25,000	1,53,000	Machinery	80,000	55,000
			Land	40,000	50,000
			Building	35,000	60,000
Total	2,30,000	2,47,000	Total	2,30,000	2,47,000

Additional Information: During the year machine costing Rs.10,000 (accumulated depreciation Rs. 3,000) was sold for Rs. 5,000. The provision for depreciation against machinery as on 1-1-2012 was Rs. 25,000 and on 31-12-2012 Rs. 40,000. Net profit for the year 1986 amounted to Rs. 45,000. You are required to prepare funds flow statement.

11. Give a detailed note on the principles and conventions of accounting.

12. Following information has been made available from the cost records of United Automobile Ltd. Manufacturing spare parts :

Direct materials Per unit :

X Rs. 8

Y Rs. 6

Direct wages

X 24 hours @ 25 paise per hour

Y 16 hours @ 25 paise per hour

Variable overheads 150% of direct wages.

Fixed overheads (total) Rs. 750

Selling price

X Rs. 25

Y Rs. 20

The directors want to be acquainted with the desirability of adopting any one of the following alternative sales mixes in the budget for in the next period.

(a) 250 units of X and 250 units of Y.

(b) 400 units of Y only

(c) 400 units of X and 100 units of Y

(d) 150 units of X and 350 units of Y

State which of the alternative sales mixes you would recommend to the management.

13. The standard mix of product is :

X 600 units at 15 paise per unit

Y 800 units at 20 paise per unit

Z 1000 units at 25 paise per unit

The consumption was :

X 640 units at 20 paise per unit

Y 960 units at 15 paise per unit

Z 840 units at 30 paise per unit

Calculate the Material variances.

14. A firm expects to have Rs. 30,000 on 1st May 1989 and requires you to prepare an estimate of the cash position during the 3 months May to July 1989. The following information is supplied to you.

Month	Sales Rs.	Purchases Rs.	Wages Rs.	Factory Expenses (Rs.)	Office Expenses (Rs.)	Selling Expenses (Rs.)
March	40,000	24,000	6,000	3,000	4,000	3,000
April	46,000	28,000	6,500	3,500	4,000	3,500
May	50,000	32,000	6,500	4,000	4,000	3,500
June	72,000	36,000	7,000	4,400	4,000	4,000
July	84,000	4,000	7,250	4,250	4,000	4,000

Other information :

(a) 25% of the sale is for cash, remaining amount is collected in the month following that for sale.

(b) Suppliers supply goods on two months credit.

(c) Delay in payment of wages and all other expenses: One month

(d) Income tax of Rs. 10,000 is due to be paid in July.

(e) Preference share dividend of 10% on Rs. 1,00,000 is to be paid in May.

15. Enumerate the factors affecting working capital requirements.

DE–4108

Sub. Code
402

DISTANCE EDUCATION

M.C.A/M.C.A(Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

INTERNET PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on Intranet Applications.

2. Define CGI. Mention the usage of CGI.

3. Bring out the significance of Scripting Language.

4. Write short notes on Shockwave.

5. What is Plug-Ins and how do you add Plug-Ins with Existing Applications?

6. Explain the VDOlive technology.

7. Define Animation and explain it.

8. Write a note on creating Dynamic Graphics?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the features of Java language with examples.

10. Discuss on PERL with examples.

11. Create dynamic page using Vbscript for mobile shopping.

12. List the different types of tags in HTML and explain it with examples.

13. Explain ActiveX control and create graphics symbols using ActiveX control.

14. Explain the different types of Web Server with relevant examples.

15. Write a detailed note on Frame and Images in Graphics.

DE–4109

Sub. Code
403

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2018.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Mention and describe the roles that may be played by an object as a participant in a link.

2. Describe the role of classes and objects in analysis and design. Suggest the metrics to know if a given class or object is well designed.

3. Describe meta model and show how to derive use cases for an application.

4. Describe the concept of parameterized classes with a sample system.

5. What is state diagram? Describe the steps to draw a state diagram for an application.

6. Give some benefits of the object model and the open issues to be addressed in effectively applying the elements of the object model.

7. Describe how to identify key abstractions and mechanisms.

8. Explain the two essential elements of a class diagram with a sample system.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give the meaning of concurrency with one example. Distinguish between heavyweight and lightweight concurrency.

10. Explain any two general approaches for classification. Give the reason for refining key abstractions.

11. Explain the overview of unified process with its four phases.

12. Explain the two essential elements of a class diagram with a sample system.

13. Define interaction diagram with a sample system. Explore the two straightforward elements that can be added to make interaction diagrams more expressive in the presence of certain complicated patterns of interaction.

14. Mention the four major elements of object model and explain any two of them.

15. List different kinds of class relationships and explain any three of them.

DE–4110

Sub. Code
404

DISTANCE EDUCATION

M.C.A. / M.C.A (Lateral) DEGREE EXAMINATION,
MAY 2018.

Fourth Semester

COMPILER DESIGN

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain about compilers, interpreters, assemblers, cross compliers.

2. Write a note on Boot strapping.

3. Explain the different phases of compiler.

4. Explain the method of constructing an NFA from a regular expression with an example.

5. Explain the role of lexical analyzer.

6. Explain construction of LALR sets of items.

7. Write a note on syntactic-phase errors.

8. What is the importance of code optimization?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the role of intermediate code generator.

10. Construct a minimum state DFA to accept all strings represented by the regular expression ((a+b) (a+b))*.

11. Explain the following

    (a) Structure of a compiler (b) Compiler-writing tools.

12. Give the model for predictive parser and explain it briefly.

13. Explain how syntax-directed translation scheme is used for generating intermediate code with a suitable example.

14. Discuss the following

    (a) Problems in code generation

    (b) Peephole optimization.

15. Discuss about synthesized attributes with suitable example.

DE–7051

Sub. Code
405

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

INTERNET PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Break up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner has to select ONE question and give it to the students by the lot system.

1. (a) Create a web page in the format of front page of a newspaper using Text link. Align the text with colors.

   (b) Write a Java program to display the different colors and fonts using Applet.

2. (a) Design and publish a web page for a college.

   (b) Write a Java program to create the frames and its controls using Applet.

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3. (a) Develop a web page for job recruitment agency in an IT industry.

   (b) Write a Java program to display the dialogue menu using applet.

———————————————— cut here ————————————————

4. (a) Develop a single page advertisement for a shop to be opened shortly.

   (b) Write a Menu-Driven Java program to perform database functions using database connection facility. The functions include display, edit, insert and delete a records from a table.

———————————————— cut here ————————————————

5. (a) Develop a picture gallery having at least 5 pages. Each of them is having several pictures with suitable information.

   (b) Write a Java applet to display any 3 images when 3 buttons in the border layout are clicked. The image should be displayed at the center.

DE–7052

Sub. Code
406

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

COMPILER DESIGN LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the students by the LOT system.

1. (a) Write a program to find keywords in a source program.

   (b) Write a program to convert infix expression into postfix notation.

---------- Cut here ----------

2. (a) Write a C program for DFA to accept the regular expression (0/1)*.

   (b) Write a C program to recognize identifier in the given source code.

---------- Cut here ----------

3. (a) Write a program to construct leftmost and rightmost derivations for the following sentences from the given grammar.

   (i) id * (id + id)

   (ii) (id + id) * (id + id)

   The grammar is

   $E \to E + T \mid T, T \to T * F \mid F, F \to (E) \mid id$.

   (b) Write a program to remove comment lines from a source program.

4. Write a program to construct LR(0) parsing table for the following grammar $S \rightarrow CC, C \rightarrow Cc \mid d$.

---------- Cut here ----------

5. (a) Write a C program to construct a canonical CR parsing table for the grammar $S \rightarrow CC, C \rightarrow Cc \mid d$.

   (b) Write a C program to recognize constant in the given source code.

---------- Cut here ----------

6. (a) Write a C program to translate the expression $(a-b)/(b+c)$ into quadruples triples and indirect triples.

   (b) Write a C program to check the production rule whether it is an operator, grammar or not $E \rightarrow EOP\ EE \rightarrow (E), E \rightarrow id, OP \rightarrow +/-/*/1$.

---------- Cut here ----------

7. (a) Write a C program to draw the transition diagram for DFA by accepting regular expression (1/0)1 +10(1/0)*.

   (b) Write a C program to find the TRAILING set for the grammar $S \rightarrow a, S \rightarrow (T), T \rightarrow T, S, T \rightarrow S$.

---------- Cut here ----------

8. (a) Write a C program to check the production whether it is a operator, grammar or not.

   $E \rightarrow E\ OPEE \rightarrow (E)$

   $E \rightarrow id$

   $OP \rightarrow +/-/*//$

   (b) Write a C program to remove new lines in the given source code.

DE–4111

Sub. Code
501

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

DISTRIBUTED COMPUTING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the Internet and Intranet technologies in Distributed Systems.

2. How would you configure a firewall to protect the Local Area Network at your institution or company? What incoming and outgoing requests should it interrupt?

3. Write short notes on Sockets.

4. Differentiate between Connectionless and Connection Oriented Protocol.

5. Bring out the significance of Kernel for a user level implementation of Threads.

6. What is Security Policy? List down the corresponding mechanisms to be followed in case of a multi user Operating System such as UNIX.

7. Explain the Mutual Exclusion Algorithm.

8. Give a brief note on simple synchronization without transactions.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write a detailed note on Software Layers in Architecture Models.

10. Explain the different types of Network with suitable diagrams.

11. Elaborate on the Client Server Communication Model.

12. Bring out the significance of Threads.

13. Give a detailed description on Cryptography Algorithm.

14. Explain in detail about the synchronizing physical clocks.

15. Write a detailed note on Locks and Locks implementation with suitable illustrations.

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DE–4112

Sub. Code
502

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2018.

Fifth Semester

WEB TECHNOLOGY

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is CSS? List out the various CSS properties.

2. What is XML? Explain XML schema with example.

3. What is Java beans? List the benefit of Java Beans

4. Explain about BDK

5. Explain the concept of Servlets with an example program

6. Explain the servlet life cycle model

7. Explain Error handling methods in JSP

8. Explain in detailed about deploying JAVA Beans in a JSP Page.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is HTML and Discuss the various HTML tags in detail

10. What is EJB's? Explain in detail.

11. Explain the following :

    (a) Handling HTTP request and responses

    (b) Cookies

12. Explain in detailed about Session Tracking and Security issues.

13. Explain JSP application design with MVC.

14. Briefly explain about JSP environment.

15. Explain how to access a database from a JSP Page with a suitable example.

DE–4113

Sub. Code
503

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

.NET FRAME WORKS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List the advantages of .Net Framework.

2. What is CLR? Elaborate.

3. Explain any five string handling functions available in .Net.

4. Explain collection classes in .Net framework.

5. Define thread. Explain the various states of thread with example.

6. List various .Net File types.

7. Describe any five controls available in .Net.

8. Describe the web services in .Net.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the architecture of .Net framework.

10. Discuss the different types of Assembly.

11. Write a note on :

    (a) Server Object types

    (b) Remoting

    (c) Threads.

12. Explain the .Net Web server Controls.

13. Explain about various Namespaces of .Net framework.

14. Explain ADO .Net features.

15. Explain the various looping statements available in .Net framework.

DE-4114

Sub. Code
504

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

DATA MINING AND WAREHOUSING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List out some impact of Data mining.
2. Explain the Knowledge discovery process.
3. What is concept hierarchy? Describe the OLAP operations in the Multidimensional data model.
4. Explain the principles of Data warehousing.
5. Differentiate Data Mart from data warehouse
6. What is meant by service level agreement?
7. List some security measures for data warehouse.
8. What is Capacity Planning?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the concepts, task and porocesses involved In data mining with examples.
10. Write detail note on : Distinguish between OLAP and OLTP.

11. Discuss in detail about the Data warehouse Architecture?

12. Distinguish between testing the data warehouse and tuning the data warehouse?

13. Discuss in detail about what are the backup and recovery procedure used in data warehouse?

14. Explain the design issues of hardware and operational design of data warehouse?

15. How genetic algorithm can be used for mining data and Explain the steps involved in KDD process.

DE–7053

Sub. Code
505

DISTANCE EDUCATION

M.C.A./M.C.A., (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

.NET LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write a VB.Net/C# Program to accept a character from console and check the case of the character.

   (b) Create table called customer with field's customer id, cus_name, address, phone_no, mail_id and then insert few customer information into customer table.

---------- Cut here ----------

2. (a) Write a VB.Ne/C# Program to accept any character from keyboard and display whether it is vowel or not.

   (b) Develop a VB.Ne/C# Program using Datagrid to display records.

3. (a) Write a VB.Net/C# Program to accept a string and convert the Case of the character.

   (b) Develop a database application to store the details of students using ADO.NET.

---------- Cut here ----------

4. (a) Write a Program to display the vacation in calendar.

   (b) Write a VB.Net/C# program for student mark sheet and generate a crystal report from an existing Database using VB.

---------- Cut here ----------

5. (a) Design a form to select image from list and display it in the picture box.

   (b) Develop a database application for Employee payroll system using ADO.Net with insert, delete and update operations.

---------- Cut here ----------

6. (a) Write a VB.Net Program to bind data using button column in data grid.

   (b) Write a Program to create an Advertisement using Ad rotator.

7. (a) Write a Program to display the selected date in the calendar control.

   (b) Create a web page with three different controls to the ASP.Net page for reserving rooms in the hotel (Three controls are button, label and a drop-down list control).

---------- Cut here ----------

8. (a) Creating simple web page using ASP.Net that has taking the user name and password as input, after the user enters the appropriate values the validation control must validates the values entered.

   (b) Write a VB.Net Program to bind data using hyperlink column in data grid.

DE–7054

Sub. Code
506

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

WEB TECHNOLOGY LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system

1. (a) To develop a html document using CSS. With font size, font face, background etc.,

   (b) Create an attractive web page of your university faculty recruitment using web graphics and proper text links with background and foreground Colors.

---------- Cut here ----------

2. (a) Write an XML document to display Customer profile.

   (b) Write an XSL style sheet and attach to the XML document and validate the document.

3. (a) Write an ASP program to prepare the Telephone bill using Javascript.

   (b) Write JSP program to display Employer Id, Employer name, Employer designation and salary using JDBC.

---------- Cut here ----------

4. (a) Writ a program to display day, month and year using servlets.

   (b) Write an ASP program to prepare Vehicle Insurance Claim Bill.

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5. (a) Write an XML document to display your favorite Freedom Fighters. Write an XSL style sheet and attach that to the XML document validate the document using XSD.

   (b) Write a JSP program to display patient Id, patient name, disease type, doctor name, fees paid, using JDBC.

---------- Cut here ----------

6. (a) Write an XML Document to Display your Students name, Date of Birth, Address and Phone Number.

   (b) Write a JSP program to prepare EB bill using java script.

7. (a) Design the static web pages required for an online book store web site.

(b) Write an ASP program to prepare student performance evaluation document using Java script.

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8. (a) Write an JSP program to collect user details and to prepare Pan Card using JavaScript.

(b) Create an HTML table for your Class time table using Row span and Col span.

DE–4115

Sub. Code
601

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

MOBILE COMMUNICATIONS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Brief note on the Motivation for Specialized MAC.

2. Explain about Wireless ATM Working Group.

3. Describe about Infrastructure and Ad-hoc Network.

4. With suitable diagram explain packet delivery from one mobile node to another node.

5. What are the Entities and Terminologies used in Mobile IP?

6. Discuss the Concept Selective Retransmission.

7. Discuss about Transmission /Timeout freezing.

8. Describe about Datagram Protocol.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Briefly describe about Digital Video Broadcasting with its Scenario and also its uses in high-speed internet access.

10. Discuss in detail about Bluetooth.

11. Explain about the importance of Location Management and Addressing Mobile Quality of Service.

12. Explain in detail about Dynamic Host Configuration Protocol (DHCP).

13. Explain with suitable diagram about the Transaction oriented TCP.

14. Explain the following :

    (a) Indirect TCP

    (b) Snooping TCP.

15. Explain about WAP System Architecture with Suitable diagram.

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DE-4116

Sub. Code
602

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2018.

Sixth Semester

MIDDLEWARE TECHNOLOGY

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss the n-tier model of client/server architecture.

2. Compare 2-tier and 3-tier models of client/server architecture.

3. Briefly discuss the JDBC data management service.

4. Describe the CORBA as interaction service.

5. Explain stateless and stateful beans with suitable example.

6. Write short notes on CMP and BMP.

7. What are the validation controls in ASP.NET? Mention uses of them.

8. How to connect web services to a data source? Explain.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the MVC architecture in detail with neat diagram.

10. With a neat diagram, discuss the DOT NET architecture.

11. Give an overview of presentation services.

12. Discuss in detail the EJB component model.

13. Write detailed notes on AdRotator and calendar control.

14. Explain the architecture of ASP.NET with neat diagram.

15. Describe HTTP and SOAP web services.

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DE–4117

Sub. Code
603

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

AGENT BASED INTELLIGENT SYSTEM

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is an intelligent agent? Write a brief note on Agents and their environments.

2. Discuss the different heuristic approaches.

3. Write a brief note on knowledge based agents.

4. Enumerate the representation language used in planning.

5. Describe the partial order planning approach with an example.

6. Explain how symbolic reasoning is measured under uncertainty.

7. What is the use of augmented grammar in natural language processing?

8. Write a brief note on the future of AI.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the structure of agents.

10. Explain in detail about the Heuristic Search techniques.

11. Discuss the knowledge engineering in first order logic.

12. Discuss the different issues in knowledge representation.

13. Explain Baye's Rule and enumerate its use.

14. Discuss in detail about the genetic algorithm.

15. Discuss the different stages involved in analyzing a sentence in natural language processing.

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DE–4118

Sub. Code
604

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

OPEN SOURCE ARCHITECTURE

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Write a note on FOSS and its trends and potential.

2. Explain about Windows System Configuration and it's utilities.

3. Compare Proxy Services Vs Printer Services.

4. How CVS supports and manage Source Code Revisions? Explain.

5. How Graphical Debuggers differs from Memory Debugging? Explain.

6. Discuss about Dynamic Shared Object Libraries.

7. List out the GUI Applications.

8. Explain about Setting up email services.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Discuss about File ownership and Permissions.

10. Explain about the concept of understanding the OS boot up process.

11. Discuss about GNU Compiler Collections and its tools.

12. Explain the methods for Installing Squid Proxy and Firewall.

13. Briefly discuss about GNU debugging tools and gdb to debug programs.

14. Explain about BASH, SED and AWK Scripting.

15. Explain the Gtk + programming with suitable example.

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DE–7055

Sub. Code
605

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

OPEN SOURCE PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by lot system.

1. (a) Write a server side PHP program for student bio data and display it using HTML table format.

   (b) Write a PHP program to upload a file to the server.

---------- Cut here ----------

2. (a) Write a PHP program to implement telephone directory that provides insertion, deletion, updation and display using MYSQL.

   (b) Write a shell program to find the details of an user name.

---------- Cut here ----------

3. (a) Write a PHP program to implement an application for shopping cart.

   (b) Write a PHP program using classes to create a table.

4. (a) Write a PHP program to create personal details using cookies.

   (b) Write a shell program to change the extension of a given file name.

---------- Cut here ----------

5. (a) Create a my sql table and execute queries to read, add, remove and modify a record from that table.

   (b) Write a PHP program to count the number of lines in a file.

---------- Cut here ----------

6. (a) Write a PHP program to implement arithmetic manipulation in one page and display the result in another page using session.

   (b) Write a simple PHP program to check that e-mails are valid.

---------- Cut here ----------

7. (a) Write a PHP program to create a Database application for Electricity Bill using MYSQL.

   (b) Write a shell program to find the details of an user session and exchange the extension of given file.

DE–7056

Sub. Code
606

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

MULTIMEDIA TOOLS LAB

(2010 Academic year onwards)

Time : Three hours

Maximum : 100 marks
Record : 5
Algorithm/Flow chart : 15
Program : 35
Execution : 35
Result : 10
Total : 100

Examiner has to choose any ONE question for each candidate by lot system.

1. (a) Implement a program in FLASH to develop a bouncing ball Animation.

   (b) Develop an animation using FLASH to create a rainfall effect.

---------- Cut here ----------

2. (a) Develop an animation to show a growing flower plant.

   (b) Implement a program to create a Maze using Maya.

3. (a) Develop an animation in Flash to implement Masking.

   (b) Implement an animation to show a flying bird in the sky.

---------- Cut here ----------

4. (a) Create a cartoon with Laughing face using FLASH.

   (b) Animate your name with different colours. Make the characters to blink.

---------- Cut here ----------

5. (a) Animate your name to create the Rotating effect in clockwise directions and display the text in reverse format.

   (b) Develop a animated banner for web page using dream viewer for a car demonstration.

---------- Cut here ----------

6. (a) Create any two different shapes animate the shapes slowly appear and disappear one after another.

   (b) Animate your own text to create the horizontal flipping effect.

---------- Cut here ----------

7. (a) Implement a program to animate the flight in a runway.

   (b) Develop a web page using dream viewer for your college.

8. (a) Implement a program in FLASH for bouncing a hail with a pad moving along with the ball.

   (b) Develop an animation in FLASH to glow a bulb with changing colours.

---------- Cut here ----------

9. (a) Develop an animation to show a flying butterfly in a garden.

   (b) Develop an animation in flash to show a moving car in the road.

---------- Cut here ----------

10. (a) Implement a program in flash for getting rain full effect.

    (b) Implement a program for animating flight with twin tower using flash.